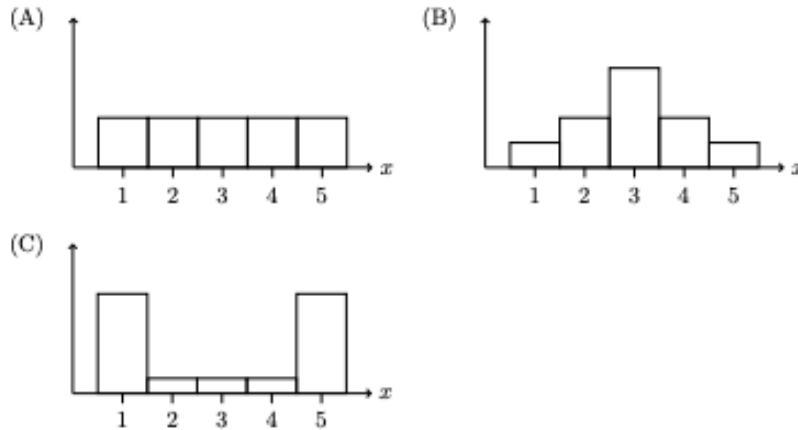
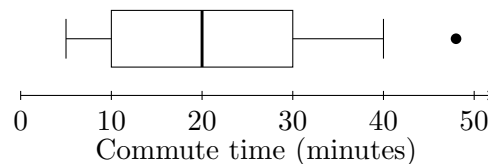


Quiz #2

- The graphs below show the probability density functions of three random variables.



- Order them by mean from smallest to largest.
 - Order them by variance from smallest to largest.
 - Order them by standard deviation from smallest to largest.
- True or False: If $\sigma^2(X) = 0$, where σ^2 is the variance of X , then X is constant.
 - The boxplot below summarizes the one-way commute times (in minutes) for a group of students.



- Approximately what is the median commute time?
 - Approximately what interval contains the middle 50% of commute times?
 - What does the point near 48 minutes indicate? What does the boxplot suggest about the shape of the distribution?
- In words, explain what this code does. How many plots are produced, and what is shown in each plot?

```
variables = ["height", "weight"]  
  
for variable in variables:  
    plt.hist(df[variable], bins=10)  
    plt.title(variable)  
    plt.show()
```